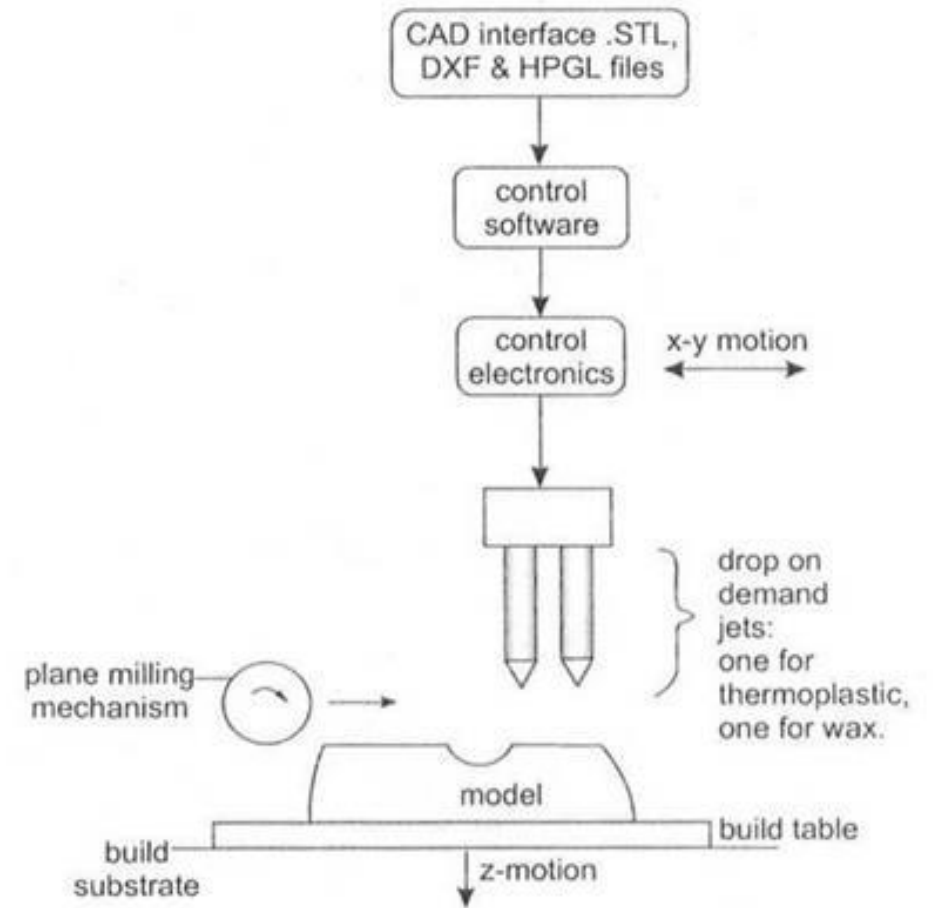
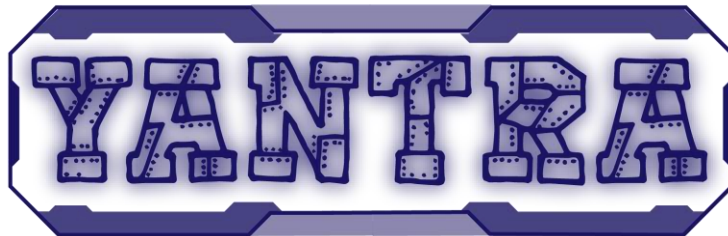


Sanders Model Maker

- The Sanders Model Maker (MM) series
- Captures the **essence of the ink-jet printing** technology.
- Builds in a **layer-by-layer** fashion, like other rapid prototyping (RP) systems.

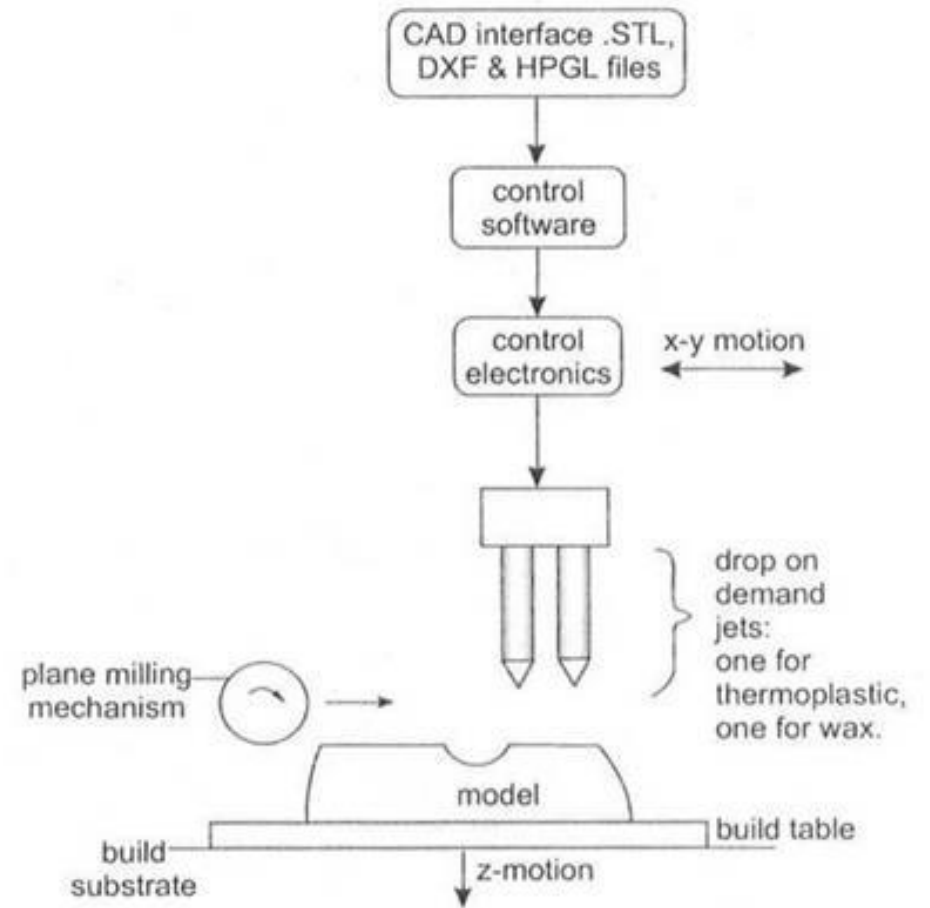


Material

- Both models use a **build** and **support material** to produce a 3-D model.
- These materials are **wax based**
 - Support having a lower melting point than the build.
- **Support material** will **melt away** leaving only the part, made of build material, During post processing.
- **Each material** has **its own heated reservoir**
 - Very sensitive to contamination, which means handling should not occur.

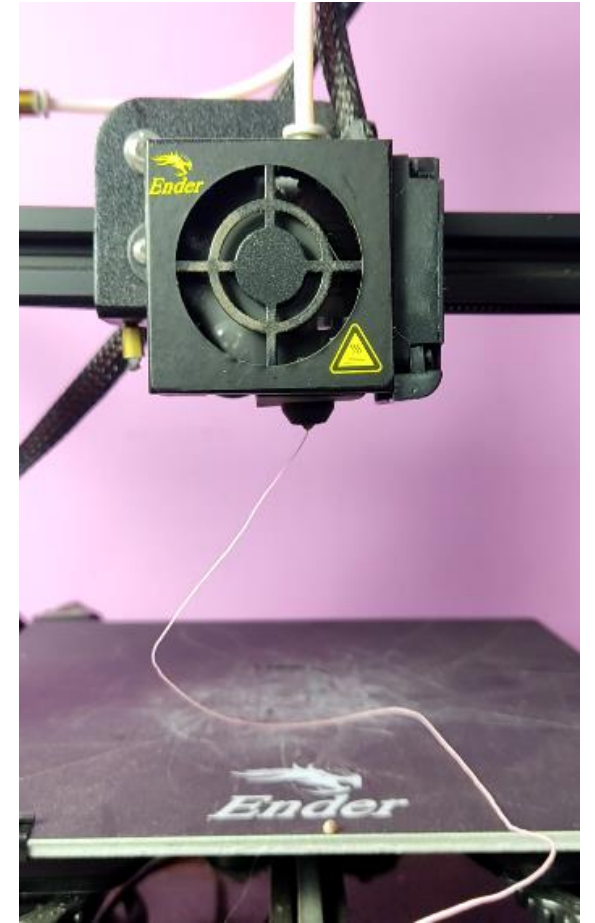
Model Makers Working

- Material is **pumped** to the feed lined by **compressed air** within the reservoir during the purge operation.
- There is an **actual siphon** that is conducted from the **reservoir** to the feed lines, to the printer-head during the model build.
- The feed lines are **heated**, as are the print-heads. This **heating** of reservoirs, feed lines, and print heads is necessary to have a **continual flow** of material.



Building a Part

- Initially, you can check the **material reservoirs** to determine if you need to add any build or support materials.
- Allow **45 minutes** for the material **to be liquefied** in the reservoir before use.
- **Mill** the original surface (dull finish) of the substrate until it has a clean, bright finish.
- This ensures that the surface is level.
- The next and most important step is to **check the jet-firing status**.
- Make sure that the proper amount of air is within the reservoir.
- Allow the jet to **purge** until you get an even flow of material into the tube container and allow it to flow for 2 to 3 seconds, then press any key to stop the purge.
- After making sure that the jets are firing properly, go back into new build, select the file you want, and **build**.



Purging in FDM:
Removing
material from
before printing

ADVANTAGES

- produce extremely fine resolution and surface finishing; as good as CNC machining
- allow designers to effectively communicate design intent to others - other designers
- The Thermo Jet printer overcomes the limitations of 2D simulations, which do not fully convey design intent and require a technical understanding of how to interpret such drawings or screen views
- The Thermo Jet printer's automated build process can supply this model in a few hours, compared to days or weeks for traditional modelmaking processes.

Sanders Model Maker

DISADVANTAGES

- Relative long build time
- After the build, the part is enclaved inside a wax block and is required some solvent to dissolve wax
- Currently, only one proprietary thermoplastic material is available which limited the application of the process
- Because of the low speed and the size of the working envelop, it is not appropriate for large parts